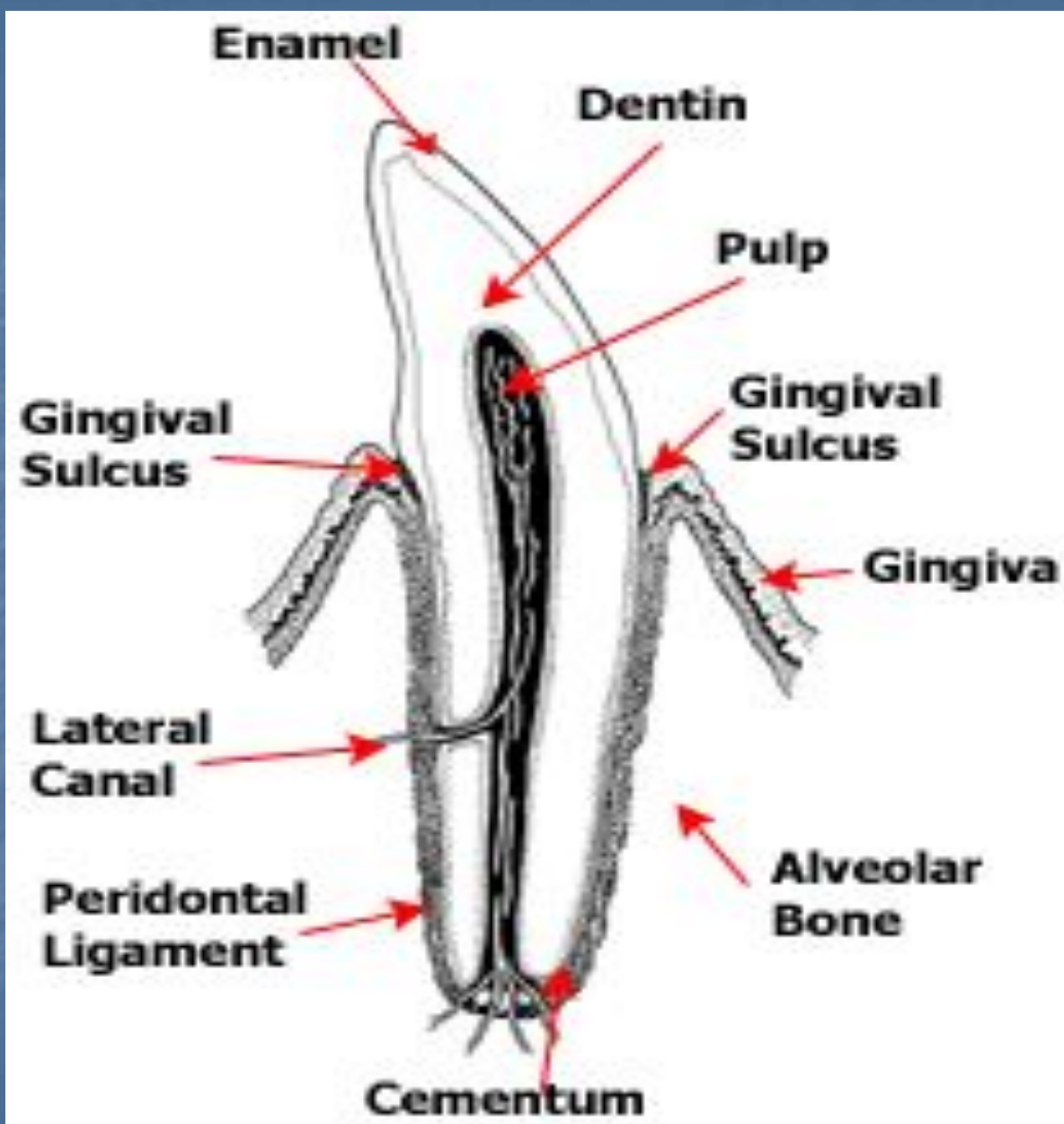


TOOTH

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Teeth

- Each tooth consists of 3 parts:
- A crown □ project beyond the gum margin.
- A root □ buried in the bony socket (alveolus).
- The region where the crown and root meet □ neck.



- Center of each tooth present an elongated cavity □ pulp cavity.
- Cavity contains loose connective tissue □ dental pulp.
- Pulp cavity extends to the apex of the root and opens via an apical foramen.
- The hard tissues of the tooth are dentine, enamel and cementum.

The hard tissues:

- The dentine surrounds the pulp cavity and forms the bulk of the tooth.
- The enamel covers the dentine of the crown.
- Cementum overlies the dentine of the root.

Soft tissues

- Include:
- The pulp□ occupying the pulp cavity.
- Peri-odontal membrane(or ligament)□ which surrounds the cementum and helps to fix the tooth firmly in socket.

Dentine

- A calcified tissue similar to bone, much harder.
- Contains larger quantities of calcium salts□ constitute 80% of its dry wt (bones 46%).
- Calcium salts are present in the form of hydroxyapatite crystals.
- The organic components(20%)consists of collagenous fibers and glycoaminoglycans.
- There are no haversian system and any cell bodies in the dentine.



- Surrounds central pulp chamber and pulp root canal.
- Calcified matrix containing collagen type 1.
- Manufactured by odontoblasts which persist and elaborate dentine for the life of tooth.

- The dentine lodges cytoplasmic processes of cells called odontoblast, bodies of which lie in the pulp cavity.
- These processes lie within narrow canals
□ dentine tubules.
- The odontoblast are dentine forming cells□ persists throughout the life.

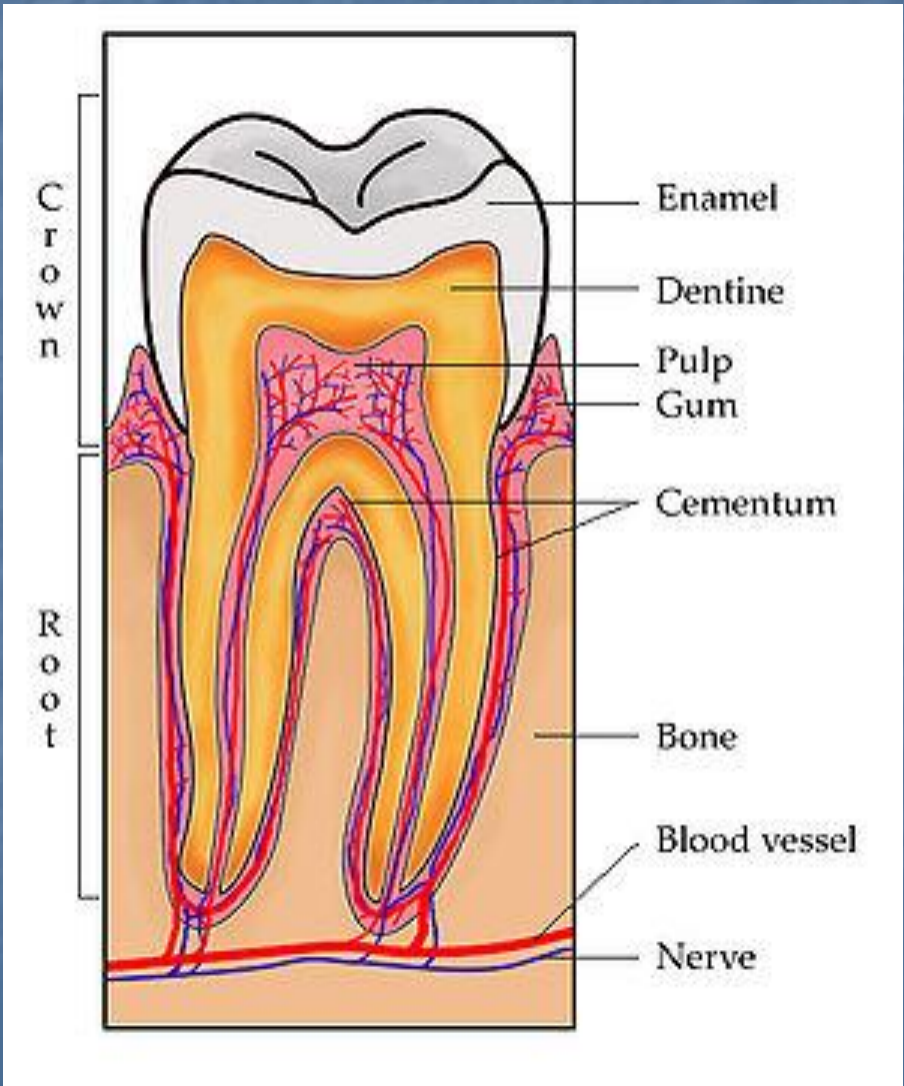
Dentine



- Odontoblast are tall columnar cells lying as single layer on the inner aspect of the dentine.
- Dentine is sensitive to heat, cold, pain and hydrogen.

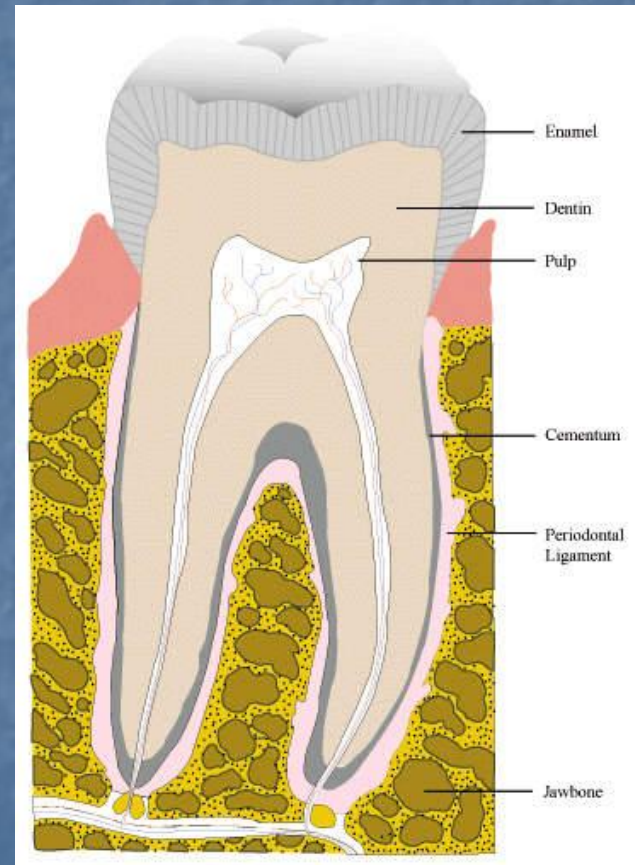
OVERVIEW

- Calcium hydroxyapatite.
- Crown-covered with enamel.
- Root-covered with cementum.
- Cervix-junction of two surface materials.



DENTAL SUPPORTS

- Periodontal ligament extends from cementum to bone, suspends tooth in alveolus.
- Gingiva covered by stratified squamous keratinized epithelium.
- Alveolar bone which is cancellous. inner layer called cribriform plate and outer layer called cortical plate.



Enamel

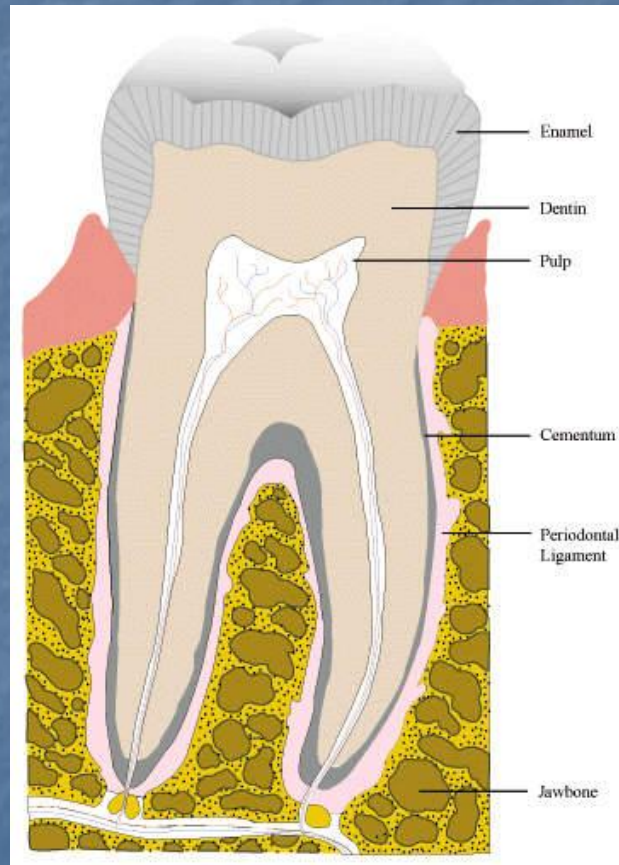
- Enamel covers the crown of the tooth.
- Hardest substance in the body.
- Ratio of inorganic and inorganic component is 98% and 2 %.
- Inorganic crystal □ apatite crystals.
- Organic component □ protein, enamelin.
- Structural unit □ enamel prism(rod).
- Bound together by interprismatic substances.

- Enamel prism are hexagonal.
- Lie perpendicular to the outer surface of dentine.
- Extend from dentine- enamel junction to the the tooth surface.
- It is ectodermal in origin.
- The other structures of tooth are derived from mesoderm.

Cementum

- Covers the dentine of the root.
- Resembles bone in composition but there are no haversian system.
- In upper half of the of the length of root□ it is devoid of cells called acellular cementum.
- Remainder of the cementum contains cell□ cementocytes.
- The cementum serves to attach the tooth to the peri-odontal membrane.

CEMENTUM



- Collagenous fibers from this membrane penetrate the cementum as Sharpey's fibers.

Pulp cavity and dental pulp

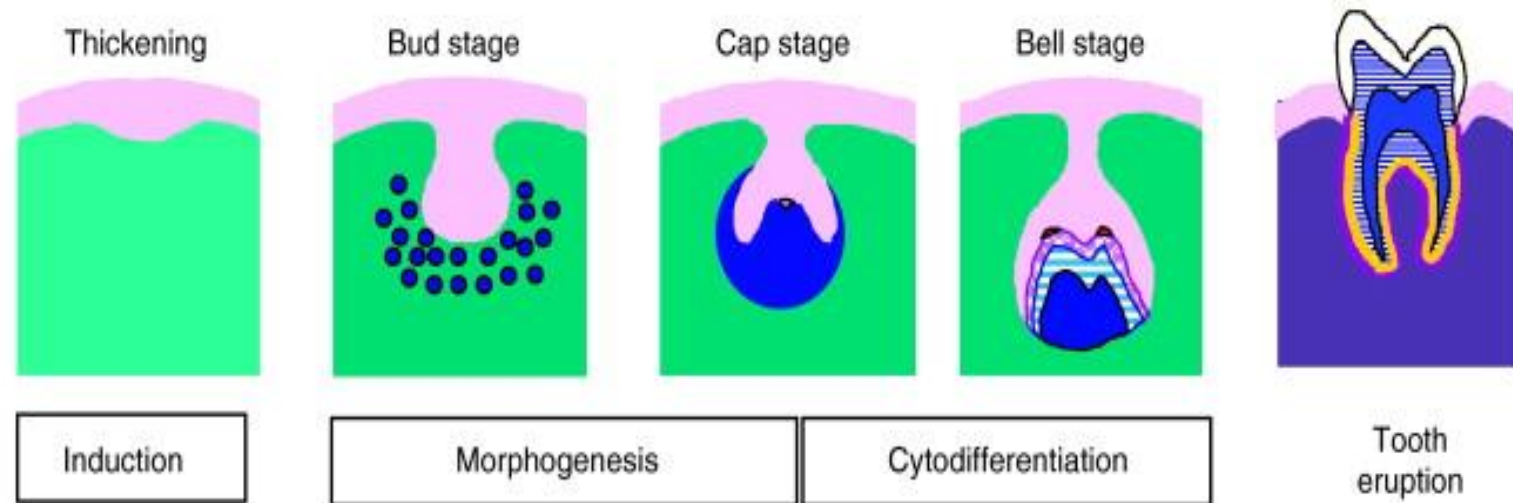
- Pulp cavity consist of upper dilated portion □ pulp chamber.
- A lower narrower portion □ pulp canal or root canal.
- Dental pulp is loose connective tissue, consists of cells and intercellular substance.
- Cell are stellate in shape with long cytoplasmic process.
- Intercellular substances consists of fine collagenous fibrils and ground substance, GAGs.
- Odontoblast lie at the periphery of the pulp in a single layer.
- Pulp also contains blood vessels and nerves of the tooth.

Development of tooth

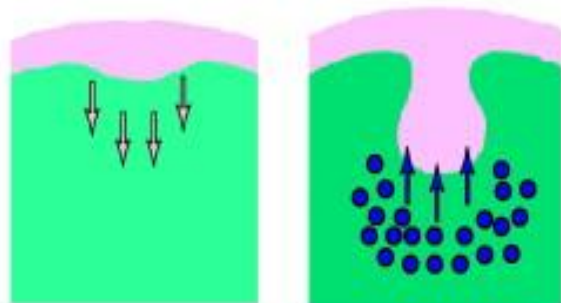
- Teeth arise from an epithelial-mesenchymal interaction b/w overlying oral epithelium and underlying mesenchyme derived from neural crest cells.
- By 6th week of development the basal layer of lining epithelium of oral cavity form a c-shaped structure, DENTAL LAMINA, along the length of upper and lower jaw.

- Lamina □ a number of DENTAL BUDS, 10 in each jaw which form primordia for the ectodermal components of teeth.
- Soon deep surface of the bud invaginates, resulting in **CAP STAGE OF TOOTH DEVELOPMENT.**
- Consists of 2 layers:
- Outer layer □ outer dental epithelium
- Inner layer □ inner dental epithelium
- Central core is called STELLATE RETICULUM.
- Mesenchyme □ originates in the neural crest cells □ in the indentation □ forms DENTAL PAPILLA.

Tooth Development



Direction of odontogenic signal



Key:

- Oral epithelium
- Neural-crest derived mesenchyme
- Condensed mesenchyme
- Primary enamel knot
- Secondary enamel knot
- Ameloblasts
- Odontoblasts
- Enamel
- Dentin
- Dental pulp
- Cementum
- Periodontal ligament
- Alveolar bone

BELL STAGE

- As dental cap grows down □ tooth takes an appearance of bell.
- Mesenchyme cells adjacent to inner dental layer □ **odontoblast** which later produce dentine.
- Layer persists throughout the life of tooth n provide **predentine**.
- Remaining cells of dental papilla □ form dental **pulp** of tooth.
- Epithelial cells of inner dental epithelium differentiate into t**Ameloblast**.

THANK YOU